EN843407 - 2568/2 - Programming Assignment #3

Where(?)wolf

Welcome to "Where(?)wolf," a captivating single-player board game set (that you can enjoy even when you have no one to play with) in the picturesque village of *Dreamlight Meadow Establishment (DME)*. In this quaint and mysterious locale, an unsettling secret haunts the community. A pack of cunning wolves has infiltrated the village, hiding in plain sight among the unsuspecting villagers.

Your mission, as the lone detective, is to unravel the identities of these elusive werewolves. The village is home to ***N***+***M*** characters, with ***N*** of them being undercover wolves and the remaining ***M*** innocent villagers. The catch? You're in the dark about who's who! To expose the wolves, you must navigate the charming Dreamlight Meadow, strategically interrogating characters, collecting clues, and deducing who among them is secretly howling at the moon.

As the game starts, ***N***+***M*** characters are randomly seated in a mysterious circle. *Interrogation Phase*, *Deduction Phase*, and *Transition Phase* are repeated until one of the *Game Ending* conditions are met.

Interrogation Phase

The characters engage in ***R*** rounds of intense interrogation. Each character possesses three strategic choices:

- Point to the opposite team
- Stay quiet
- Point to the same team. (This choice becomes another ‘Stay quiet’ if the character is the only remaining member of the team)

If a character selects to point at a team, one of the characters in that team (excluding themselves) will be selected with equal probability.

The Wolves and Villagers teams each have distinct probabilities for their actions:

- **Wolves**: Probability of pointing to the opposite team (***W_o***), staying quiet (***W_q***), and pointing to the same team (***W_s***).
- **Villagers**: Probability of pointing to the opposite team (***V_o***), staying quiet (***V_q***), and pointing to the same team (***V_s***).

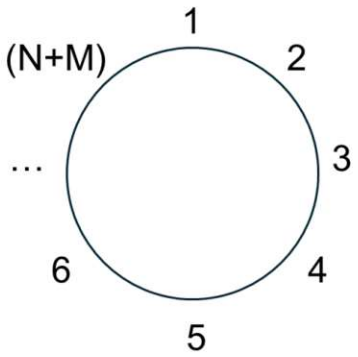
Deduction Phase

In this phase, you, as the detective, utilize your deduction skills to select one character for removal from the game.

Transition Phase

In this phase, the chosen character unveils their role before leaving the circle, either admitting to being a wolf or affirming their innocence as a villager.

After that, without revealing roles, each wolf character has a chance to transfer his wolf role to the character on his right. This process is done in the order of seating location.



- Starting with the character seated at location 1, if the character is a wolf, there is a chance, denoted as ***T***, that he will swap roles with the character to their right (location **N+M**).
- The process is repeated for the character at location 2, if he is a wolf, there is a ***T*** chance that he will swap roles with the character to their right (location 1).
- The process continues until the character at location **N+M**.

Game Ending conditions

The game relentlessly repeats this cycle until one of two decisive outcomes:

- **No Wolves Left**: The wolves are successfully exposed and eliminated, leaving the remaining characters as innocent villagers. The village is safe, and your detective skills have triumphed.
- **No Villagers Left**: The werewolves have successfully deceived and eliminated all villagers, plunging Dreamlight Meadow into darkness.

Your Task:
You will use what you learned in class (probably related to Hidden Markov Models) to play the ***Where(?)wolf*** game. Your task is to implement a class ***Wherewolf*** with the functions ***interrogation***(), ***deduction***(), and ***transition***().

The constructor of the class takes the following parameters:

- ***N*** - The number of wolf characters
- ***M*** - The number of villager characters
- **R** - The number of rounds in the Interrogation Phase
- **W_o** - The probability that a wolf character points at the opposite team
- **W_q** - The probability that a wolf character stays quiet
- **W_s** - The probability that a wolf character points at the same team
- **V_o** - The probability that a villager character points at the opposite team
- **V_q** - The probability that a villager character stays quiet
- **V_s** - The probability that a villager character points at the same team
- **T** - The probability that a wolf character transfers his role to another.

The ***interrogation***() function takes one argument ***hint***, which is an array of (***N***+***M***) integers indicating responses of each character. (The value of zero means the character stays quiet, or the character is already removed from the game.)

The ***deduction***() function takes no argument but expects a returned value, which is the answer of your deduction (the location of your suspect.) Your answer must be between 1 to (N+M) and must not be an empty seat (the character was already removed in previous iteration); otherwise, the game ends and you lose.

The ***transition***() takes one argument, ***revealed***, revealing the role of the character that you chose to remove. You may also implement any code to represent the role transfer process.

Workflow of the Program:

The game will be played through a series of function calls, representing each phase/round of the game. Below is the workflow of your program. You have **five seconds** to return the answer in each round of deduction.

```
                mygame = Wherewolf(N,M,R,W_o,W_q,W_s,V_o,V_q,V_s,T)     // Constructor
                mygame.interrogation(hint)                              // Interrogation - 1st round
                mygame.interrogation(hint)                              // Interrogation - 2nd round
within 5 seconds ...
                mygame.interrogation(hint)                              // Interrogation - R th round
                answer = mygame.deduction()                             // Provide your deduction
                mygame.transition()                                     // Transition Phase
                mygame.interrogation(hint)                              // Interrogation - 1st round
within 5 seconds mygame.interrogation(hint)                             // Interrogation - 2nd round
                ...
                mygame.interrogation(hint)                              // Interrogation - R th round
                answer = mygame.deduction()                             // Provide your deduction
                mygame.transition(revealed)                             // Transition Phase
                ...
```

Starting Code:

You are provided with the following starting code:

- ***run.py*** An example judge program.
- ***Wherewolf.py*** Your starting file. **DO NOT CHANGE NAME**

Grading Criteria:

Your program will be intensively run on multiple scenarios. Your score is determined by the number of remaining villagers when the games conclude. **Any timed out or compiled-error runs result in zero points.**

Submission:

As automatic scripts will be used for grading, to ensure that your submission will be graded, please follow the instructions below:

- Compile the following files into **one .zip file**
  - Your ***Wherewolf.py*** file (do not rename the file), with your name(s) and ID at the top.
    - Please make sure your file does not import functions from other local files; otherwise, it will not run during the grading.
  - README file with your name and ID at the top
- Name your zipped file as ***<ID>.zip*** (without a hyphen)***.*** For example, ***1234567890.zip .***

Important Notes:

- You are encouraged to discuss with your friends about how to solve this assignment, but please do NOT share your code, or tell others what to type.
- You may not use a code generator tool, such as ChatGPT, GitHub Copilot, or other similar software. The purpose of this assignment is to develop your understanding and implementation skills in search algorithms; relying on automated code generation undermines the learning process and creative problem-solving skills.

If you have any questions, please feel free to contact the instructor
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